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**Prog part 1**

## Section B — API Endpoint Plan

The RaceDay API will provide endpoints for authentication, user profiles, events, categories, event enrolments and results. Access to endpoints will be controlled according to the user's role.

| HTTP Method | Route              | Description                                          | Role Required          | Request Body                                               | Expected Response                                                   |
|-------------|--------------------|------------------------------------------------------|------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| POST        | /api/auth/register | Register a new user based on their selected role.    | Public                 | First name, last name, email, password, role, phone number | <b>201 Created</b> – User registered successfully.                  |
| POST        | /api/auth/login    | Authenticate a user and log them into the system.    | Public                 | Email, password                                            | <b>200 OK</b> – Login successful and authentication token returned. |
| GET         | /api/users/profile | View the profile information of the logged-in user.  | Any authenticated user | None                                                       | <b>200 OK</b> – User profile details returned.                      |
| PUT         | /api/users/profile | Update the logged-in user's own profile information. | Any authenticated user | First name, last name, phone number                        | <b>200 OK</b> – Profile updated successfully.                       |
| GET         | /api/events        | View all available events.                           | Any authenticated user | None                                                       | <b>200 OK</b> – List of events returned.                            |
| GET         | /api/events/{id}   | View the details of a specific event.                | Any authenticated user | None                                                       | <b>200 OK</b> – Event details returned.                             |
| POST        | /api/events        | Create a new running, walking or cycling event.      | Organiser              | Name, description, date, location, distance, event type    | <b>201 Created</b> – Event created successfully.                    |

|        |                                  |                                                                       |                        |                                                                      |                                                        |
|--------|----------------------------------|-----------------------------------------------------------------------|------------------------|----------------------------------------------------------------------|--------------------------------------------------------|
| PUT    | /api/events/{id}                 | Update an existing event.                                             | Organiser              | Updated event details                                                | <b>200 OK</b> – Event updated successfully.            |
| DELETE | /api/events/{id}                 | Delete an existing event.                                             | Organiser              | None                                                                 | <b>204 No Content</b> – Event deleted successfully.    |
| GET    | /api/events/{eventId}/categories | View all categories available for a specific event.                   | Any authenticated user | None                                                                 | <b>200 OK</b> – List of categories returned.           |
| POST   | /api/events/{eventId}/categories | Create an age or distance category for an event.                      | Organiser              | Category name, category type, description, maximum participants, fee | <b>201 Created</b> – Category created successfully.    |
| PUT    | /api/categories/{id}             | Update an existing event category.                                    | Organiser              | Updated category details                                             | <b>200 OK</b> – Category updated successfully.         |
| DELETE | /api/categories/{id}             | Delete an event category.                                             | Organiser              | None                                                                 | <b>204 No Content</b> – Category deleted successfully. |
| POST   | /api/enrolments                  | Enrol a participant into an event by selecting a category.            | Participant            | Event ID, category ID                                                | <b>201 Created</b> – Enrolment created successfully.   |
| GET    | /api/enrolments/my               | View all events and categories the logged-in participant has entered. | Participant            | None                                                                 | <b>200 OK</b> – Participant's enrolments returned.     |
| GET    | /api/events/{eventId}/enrolments | View all participant enrolments for an organiser's event.             | Organiser              | None                                                                 | <b>200 OK</b> – List of event enrolments returned.     |

|      |                                      |                                                                            |             |                                 |                                                    |
|------|--------------------------------------|----------------------------------------------------------------------------|-------------|---------------------------------|----------------------------------------------------|
| POST | /api/enrolments/{enrolmentId}/result | Capture a participant's finish time and finishing position after an event. | Organiser   | Finish time, finishing position | <b>201 Created</b> – Result recorded successfully. |
| GET  | /api/results/my                      | View the logged-in participant's personal results.                         | Participant | None                            | <b>200 OK</b> – Participant's results returned.    |
| PUT  | /api/results/{id}                    | Update an existing participant result.                                     | Organiser   | Finish time, finishing position | <b>200 OK</b> – Result updated successfully.       |

## Role Definitions

- **Public:** The endpoint can be accessed without logging in.
- **Any authenticated user:** The endpoint requires the user to be logged in and can be accessed by either an Organiser or Participant.
- **Organiser:** Only users with the Organiser role can access the endpoint.
- **Participant:** Only users with the Participant role can access the endpoint.

## API Design Notes

The API will use standard HTTP methods to perform operations on the RaceDay system. GET will be used to retrieve information, POST to create new records, PUT to update existing records, and DELETE to remove records.

Authentication will be required for protected endpoints. Participants will only be able to manage their own profiles, enrolments and results, while Organisers will be responsible for managing events and categories and recording participant results.